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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
Rat pancreatic islets were isolated from 250 to 300 g adult male Lewis rats by collagenase digestion, separated by Ficoll gradient, handpicked under a stereo microscope and cultivated according to Ref. 9.
Subject Demographics
Age: not required.
Weight: not required.
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
The standards and each sample were determined in technical triplicates and each sample was normalized to the mean cycle threshold (CT) from three tested stable reference genes.
Type: The standards and each sample were determined in technical triplicates and each sample was normalized to the mean cycle threshold (CT) from three tested stable reference genes.

Number: The standards and each sample were determined in technical triplicates and each sample was normalized to the mean cycle threshold (CT) from three tested stable reference genes.
Cell Line Authentication
not detected.

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
To visualize AQP8, the V5 tag was stained by the primary V5 antibody (1:400) (E10V4RR, Invitrogen, Carlsbad, CA, USA) in 0.1 BSA in PBS0.1 TritonX using a Cy3-labelled secondary antibody (1:200) (Dianova, Hamburg, Germany) in 0.1 BSA in PBS0.1 TritonX.	V5		
To determine colocalization, the cells were simultaneously antibody-stained either with Con A (Concanavalin A) conjugated with Alexa Flour 488 (C11252, Thermo Fisher Scientific) for 1 h at RT after fixation and before permeabilization for the plasma membrane.	C11252		
All other immunostainings were performed after the permeabilization and blocking procedure with either the COX IV (cytochrome c oxidase subunit 4) antibody (1:1,000) (4844 Cell Signaling Technology) for the mitochondria, or with ABCD3 (1:200) ((F1):sc-514728, Santa Cruz Biotechnology, Dallas, Tx, USA) for the peroxisomes, with ERO1LB (1:100) (11261-2-AP, Thermo Fisher Scientific) for the ER (Endoplasmic reticulum), Giantin (1:1000) (PRB-114C, Covance, Princeton, NJ, USA) for the Golgi apparatus, or with Insulin (1:100) (ab7842, abcam, CA, UK).	ABCD3		
	ERO1LB		
	Giantin		
	Insulin		
Subsequently the membrane was blocked by 1 milk for 1 h at room temperature and antibody staining was performed overnight using an AQP8 antibody (1:2500) (Cat AQP81-A, alpha diagnostic, San Antonio, TX, USA) in 1 milkPBS0.1 Tween20.	AQP8		
As reference gene COX IV (cytochrome c oxidase subunit 4) antibody (1:1,000) (4844 Cell Signaling Technology) in 5 milkTBS0.1 Tween20 was used.	cytochrome c oxidase subunit 4		
PEG6000 was used to precipitate the antibody-bound insulin.	antibody-bound insulin		
Experimental Models: Cell Lines			

2 Material and methods 2.1 Cell culture The rat insulinoma-derived cell line RINm5F (RRID: CVCL_0501 ATCC CRL-11605, Manassas, VA, USA) was cultured according to Ref. 30.	RINm5F	ATCC	(ATCC Cat# CRL-11605, RRID:CVCL_0501)(link)
The rat insulinoma-derived cell line INS-1E (RRID:CVCL_0351) 31 was cultured in RPMI 1640 medium (Roswell Park Memorial Institute) supplemented with 10 mM glucose, 10 (vv) fetal calf serum, 10 mM HEPES, 1 mM pyruvate, 1 mM l-glutamine and 50 mM 2-mercaptoethanol.	INS-1E	AddexBio	(AddexBio Cat# C0018009/4675, RRID:CVCL_0351)(link)
RINm5F cells were transduced with LentiCRISPRv2-mCherry-sgAQP8 lentivirus with a MOI of 0.2 .	RINm5F		
This AQP8exo cDNA was subcloned into the restriction sites BamHIEcoRI of the Tet-On regulated pLVX-TRE3G vector system (TakaraBio, Kusatsu, Japan) and transduced via lentiviruses into RINm5F Tet3G cells that express the Tet-On 3G transactivator.	Tet3G		
Recombinant DNA			
2.4 AQP8 overexpression and KD generation AQP8 cDNA from rat colon RNA origin (accession : BC081812.1) was amplified by PCR (forward primer: rAQP8-fw - ATTAGAATTCATGTCTGGGGAGCA GACG reverse primer: rAQP8-rv - CTTCTAGATCACCTCGACTTTAGA ATCAGGC) into the EcoRIXbaI sites of the lentiviral transfer plasmid pLVX-IRES-Neo (Takara Bio Europe , Saint-Germain-en-Laye , France) and sequenced.	pLVX-IRES-Neo		
The DNA coding for the sgRNA (CGTAGGTCCATACTACACATCGG) was cloned into the lentiviral vector LentiCRISPRv2-mCherry (kind gift from Agata Smogorzewska , Addgene , Watertown , MA , USA plasmid 99154 http:n2t.netaddgene:99154 RRID: Addgene_99154) using BsmBI digest followed by the original LentiCRISPRv2 cloning protocol (https:media.addgene.orgdataplasmids attachment_B3xTwla0bkYD.pdf)	525296152961-		RRID:Addgene_99154(link)
This AQP8exo cDNA was subcloned into the restriction sites BamHIEcoRI	pLVX-TRE3G		

of the Tet-On regulated pLVX-TRE3G vector system (TakaraBio, Kusatsu, Japan) and transduced via lentiviruses into RINm5F Tet3G cells that express the Tet-On 3G transactivator.			
Respective control cells for the AQP8 overexpression were RINm5F cells termed mock were infected with an empty pLVX-Neo vector providing the neomycin resistance.	pLVX-Neo		
The PCR fragment was restricted with NheI and EcoRI endonucleases and cloned in N-terminal fusion with mCherry in a pcDNA3.1 vector.	pcDNA3.1		Suggestion: RRID:Addgene_100544(link)
Software and Algorithms			
To quantify colocalization , we used JACoP (just another Colocalization Plugin) 32 from Fiji to quantify the Pearsons correlation coefficient of 3D images .	Fiji		Suggestion: Fiji (RRID:SCR_002285)(link)
Tested clones were sequenced (GATC Service by Eurofins Genomics, Ebersberg, Germany) and analyzed by the software ContigExpress (Invitrogen) to identify mutations.	ContigExpress		
For acquisition control the CellSens Software or Metafluor Software (Molecular Devices, San Jose, CA, USA) was used and obtained over time fluorescence intensities were analyzed (after background subtraction) by using the GraphPad Prism 5 software (San Diego, CA, USA).	CellSens		
	GraphPad Prism		
After recording the HyPer ratio, the curves were analyzed using a custom code in MATLAB R2017a to fit the curves and calculate the first derivative and inflection point and thus the maximal slope of each curve (Supplementary Fig. 5).	MATLAB		Suggestion: MATLAB, RRID:SCR_001622(link)

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
To quantify colocalization , we used JACoP (just another Colocalization Plugin) 32 from Fiji to quantify the Pearsons correlation coefficient of 3D images .	Pearsons correlation
Statistical analyses were performed by GraphPad PRISM 5 (San Diego, CA, USA) using one-way or two-way ANOVA followed by Bonferroni and two-tailed unpaired Students t-test.	ANOVA
	Bonferroni
	t-test
Oligonucleotides	
2.4 AQP8 overexpression and KD generation AQP8 cDNA from rat colon RNA origin (accession : BC081812.1) was amplified by PCR (forward primer: rAQP8-fw - ATTAGAATTCATGTCTGGGGAGCA GACG reverse primer: rAQP8-rv - CTTCTAGATCACCTCGACTTTAGA ATCAGGC) into the EcoRIXbaI sites of the lentiviral transfer plasmid pLVX-IRES-Neo (Takara Bio Europe , Saint-Germain-en-Laye , France) and sequenced.	CTTCTAGATCACCTCGACTTTAGA ATCAGGC
The DNA coding for the sgRNA (CGTAGGTCCATACTACACATCGG) was cloned into the lentiviral vector LentiCRISPRv2-mCherry (kind gift from Agata Smogorzewska , Addgene , Watertown , MA , USA plasmid 99154 http:n2t.netaddgene:99154 RRID: Addgene_99154) using BsmBI digest followed by the original LentiCRISPRv2 cloning protocol (https:media.addgene.orgdataplasmids52529 6152961-attachment_B3xTwla0bkYD.pdf)	CGTAGGTCCATACTACACATCGG

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